

SANTIAGO PUDAHUEL, Chile

WMO: 855740

Lat: 33.3924S

Lon: 70.7943W

Elev: 1578

StdP: 13.88

Time Zone: -4.00 (CHL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	30.5	32.7	24.8	20.0	52.6	27.2	22.4	48.0	16.5	51.6	14.1	52.9	2.4	0	0.484

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	29.2	89.8	62.6	87.8	62.4	85.9	62.3	65.8	84.4	64.8	83.4	63.8	82.3	14.4	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.6	75.1	72.1	57.0	73.5	71.6	55.6	69.9	70.0	31.4	84.6	30.5	83.5	29.7	82.5	70.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.0	17.5	15.8	DB	26.6	94.7	2.3	2.0	25.0	96.2	23.6	97.3	22.4	98.5	20.7	99.9	(4)
(5)			WB	25.9	67.7	2.1	1.3	24.3	68.6	23.1	69.4	21.9	70.1	20.3	71.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	59.5	70.7	69.7	66.5	59.7	53.7	49.0	48.1	50.6	54.5	59.3	64.2	68.2	(6)	
	DBStd	8.93	3.21	3.34	3.60	4.31	4.36	4.27	4.10	4.74	4.51	4.44	4.16	3.69	(7)	
	HDD50	231	0	0	0	1	15	68	86	50	11	0	0	0	(8)	
	HDD65	2572	2	4	25	165	351	479	525	447	314	185	62	13	(9)	
	CDD50	3681	642	550	513	290	129	38	26	68	147	287	427	564	(10)	
	CDD65	550	179	135	73	5	0	0	0	0	0	7	39	112	(11)	
	CDH74	8581	2278	1749	1259	321	47	1	2	28	89	299	848	1660	(12)	
(13)	CDH80	2702	840	610	377	57	4	0	0	2	11	54	216	531	(13)	
(14)	Wind	WSAvg	6.0	8.4	7.7	6.5	5.1	4.1	3.8	3.8	4.4	5.4	6.5	7.7	8.5	(14)
(15) Precipitation	PrecAvg	11.6	0.0	0.0	0.2	0.7	1.8	3.1	2.2	1.8	1.0	0.4	0.2	0.1	(15)	
	PrecMax	26.3	0.2	0.5	0.8	4.1	5.6	9.6	7.3	4.3	4.2	2.2	2.0	0.7	(16)	
	PrecMin	4.3	0.0	0.0	0.0	0.0	0.3	0.2	0.4	0.2	0.2	0.0	0.0	0.0	(17)	
	PrecStd	5.0	0.0	0.1	0.2	1.0	1.5	2.6	1.5	1.4	1.0	0.6	0.4	0.2	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.3	91.6	90.2	85.4	79.0	71.8	71.9	77.6	80.9	85.7	89.3	91.3	(19)	
		MCWB	63.4	63.0	61.8	59.7	56.2	53.4	54.3	57.0	58.1	60.7	60.3	62.4	(20)	
	2%	DB	89.7	88.5	87.1	81.1	73.8	67.7	66.8	71.8	76.6	80.7	85.1	88.0	(21)	
		MCWB	63.3	63.0	61.5	58.6	54.9	52.0	52.5	54.7	56.9	58.8	60.6	62.4	(22)	
	5%	DB	87.5	86.2	84.3	77.9	69.8	63.0	62.6	66.6	71.8	77.2	82.4	85.7	(23)	
		MCWB	63.0	63.0	61.3	57.9	54.3	51.3	50.8	53.3	55.9	57.9	60.1	62.2	(24)	
	10%	DB	84.6	84.1	81.2	73.7	66.0	59.1	58.7	62.2	67.6	73.5	79.0	82.7	(25)	
		MCWB	62.9	62.7	60.8	57.0	53.6	50.7	49.7	51.5	54.5	56.7	59.4	61.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.2	67.4	65.7	62.7	59.1	56.9	55.9	58.8	60.8	62.9	64.2	66.4	(27)	
		MCDB	87.5	86.3	83.4	78.6	70.4	65.1	68.5	73.3	74.8	80.1	83.1	85.1	(28)	
	2%	WB	65.7	65.8	63.9	60.7	57.2	54.4	53.7	56.1	58.5	60.4	62.4	64.7	(29)	
		MCDB	84.8	84.0	82.0	75.6	68.3	61.9	63.5	69.0	72.5	76.9	80.9	83.6	(30)	
	5%	WB	64.6	64.6	62.8	59.1	55.7	52.8	51.9	54.0	56.7	58.8	61.1	63.4	(31)	
		MCDB	83.8	82.7	80.6	73.2	66.3	60.1	59.8	64.0	69.8	74.3	79.2	82.3	(32)	
	10%	WB	63.5	63.4	61.5	57.7	54.2	51.2	50.3	52.3	54.8	57.2	59.8	62.2	(33)	
		MCDB	82.6	81.3	78.5	70.9	63.6	57.9	57.1	60.8	65.6	71.5	77.2	80.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	29.2	29.4	29.7	26.9	24.3	21.4	21.9	22.6	23.6	25.5	28.3	29.1	(35)	
		MCDBR	32.5	32.5	34.5	35.3	33.7	30.6	31.3	33.3	34.3	33.8	33.7	32.9	(36)	
		MCWBR	12.7	13.0	14.0	16.0	17.2	17.2	18.0	18.0	17.3	15.2	14.1	13.4	(37)	
	5% WB	MCDBR	29.4	29.9	30.9	29.6	27.5	23.8	25.9	28.5	30.6	30.6	30.7	30.3	(38)	
		MCWBR	12.3	12.7	13.4	14.3	14.7	14.1	15.6	16.0	16.2	14.6	13.8	13.1	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.331	0.332	0.318	0.317	0.309	0.298	0.298	0.309	0.326	0.317	0.309	0.320	(40)	
	taud		2.462	2.472	2.502	2.493	2.512	2.569	2.530	2.488	2.419	2.460	2.500	2.479	(41)	
	Ebn at Noon		319	312	305	288	273	269	275	286	298	314	325	324	(42)	
	Edh at Noon		37	36	32	29	25	23	24	29	35	36	36	37	(43)	
(44) All-Sky Solar Radiation	RadAvg		2704	2433	1993	1390	890	684	754	978	1410	1893	2412	2711	(44)	
	RadStd		132	89	108	134	102	95	80	119	141	143	140	102	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.86	+0.95	N/A	+1.15	N/A	N/A	N/A	-151	+337	+189
(47)	Regional (3 neighbors)	+0.68	N/A	-2.11	+1.04	N/A	N/A	N/A	N/A	+247	+167

Nomenclature: See separate page